

21. ECB, CBC and CFB – Padding

```
... Enter Plain Text: HELLO

Original Length: 5 bytes
Padded Length: 8 bytes
Padded Data: b'HELLO\x80\x00\x00'
```

22. CBC Encryption and Decryption using S-DES

```
... Enter 16-bit Plaintext: 00000011000111
Enter 8-bit Key: 1001001
Enter 8-bit IV: 10011100

Ciphertext: 0000110100000
Decrypted: 0000001000111
```

23. Counter Mode

```
... Enter 24-bit Plaintext: 000011011011001101010011
Enter 8-bit Key: 01111101

Ciphertext: 011100001100111100101100
Decrypted: 000011011011001101010011
```

24.rsa private key

```
▼ ... p = 59  
      q = 61  
      phi(n) = 3480  
      Public Key = ( 31 , 3599 )  
      Private Key = ( 3031 , 3599 )
```

25. RSA – Plaintext Has a Common Factor with n

If a plaintext block has a common factor with n, then **yes, this can help**.

```
... Enter n: 3233  
    Enter plaintext block: 53  
  
    gcd(M, n) = 53  
    Common factor found!  
    Factor of n = 53  
    Other factor = 61  
    RSA modulus can be factored.
```